

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 23
Design of FIR Filters Using Frequency Sampling

III B.Tech EEE — Core Exam Preparation Guide

1 Core Concept of Frequency Sampling

1.1 Basic Principle

Instead of truncating an infinite impulse response in the time domain (windowing), the **Frequency Sampling Method** directly discretizes the desired continuous frequency response $H_d(e^{j\omega})$ at N uniformly spaced points on the unit circle:

$$\omega_k = \frac{2\pi k}{N}, \quad k = 0, 1, \dots, N-1 \quad (1)$$

The discrete frequency samples are:

$$H[k] = H_d(e^{j\omega})|_{\omega=\frac{2\pi k}{N}} = |H[k]|e^{j\theta_k} \quad (2)$$

The causal FIR impulse response $h[n]$ ($0 \leq n \leq N-1$) is computed via the N -point **Inverse Discrete Fourier Transform (IDFT)**:

$$h[n] = \frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{j\frac{2\pi kn}{N}}, \quad 0 \leq n \leq N-1 \quad (3)$$

By definition, the reconstructed filter response $H(e^{j\omega})$ passes **exactly** through $H[k]$ at all N grid frequencies:

$$H\left(e^{j\frac{2\pi k}{N}}\right) \equiv H[k], \quad \forall k \in [0, N-1] \quad (4)$$

2 Linear-Phase Constraints on Frequency Samples $H[k]$

For the synthesized impulse response $h[n]$ to be real-valued and satisfy linear phase ($h[n] = h[N-1-n]$), the frequency samples must satisfy:

1. Magnitude Symmetry (Even Symmetry):

$$|H[N-k]| = |H[k]|, \quad k = 1, 2, \dots, N-1 \quad (5)$$

2. Phase Angle Assignment (Strict Linear Phase):

$$\theta_k = \angle H[k] = -\tau\omega_k = -\left(\frac{N-1}{2}\right) \frac{2\pi k}{N} \quad (6)$$

3. DC Bin Property:

$$\theta_0 = 0 \implies H[0] \in \mathbb{R}^+ \quad (\text{Strictly real and non-negative}) \quad (7)$$

4. Nyquist Bin Property (for N even): At $k = N/2$ ($\omega = \pi$), Type 2 filters possess an obligatory zero:

$$H[N/2] = 0 \quad (8)$$

3 Derivation of Real-Valued IDFT Synthesis Formulas

3.1 Step-by-Step Derivation for Type 1 Filters (N Odd)

Substituting the polar form $H[k] = |H[k]|e^{j\theta_k} = |H[k]|e^{-j\frac{2\pi k}{N}\tau}$ into the standard IDFT definition:

$$\begin{aligned} h[n] &= \frac{1}{N} \sum_{k=0}^{N-1} H[k] e^{j\frac{2\pi kn}{N}} = \frac{1}{N} \sum_{k=0}^{N-1} |H[k]| e^{-j\frac{2\pi k}{N}\tau} e^{j\frac{2\pi kn}{N}} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} |H[k]| e^{j\frac{2\pi k}{N}(n-\tau)} \end{aligned} \quad (9)$$

For odd N , we partition the summation into the DC term ($k = 0$), the positive-frequency terms ($1 \leq k \leq \frac{N-1}{2}$), and the conjugate negative-frequency terms ($\frac{N+1}{2} \leq k \leq N-1$):

$$h[n] = \frac{1}{N} \left[H[0] + \sum_{k=1}^{(N-1)/2} |H[k]| e^{j\frac{2\pi k}{N}(n-\tau)} + \sum_{k=(N+1)/2}^{N-1} |H[k]| e^{j\frac{2\pi k}{N}(n-\tau)} \right] \quad (10)$$

In the second summation, we perform the change of variable $k = N - m$. As k ranges from $\frac{N+1}{2}$ to $N-1$, m ranges from $\frac{N-1}{2}$ down to 1:

$$\begin{aligned} \sum_{k=(N+1)/2}^{N-1} |H[k]| e^{j\frac{2\pi k}{N}(n-\tau)} &= \sum_{m=1}^{(N-1)/2} |H[N-m]| e^{j\frac{2\pi(N-m)}{N}(n-\tau)} \\ &= \sum_{m=1}^{(N-1)/2} |H[m]| e^{j2\pi(n-\tau)} e^{-j\frac{2\pi m}{N}(n-\tau)} \end{aligned} \quad (11)$$

For odd N , the delay $\tau = \frac{N-1}{2}$ is an integer. Since $n \in \mathbb{Z}$, the difference $(n - \tau)$ is strictly an integer, which gives $e^{j2\pi(n-\tau)} = \cos(2\pi(n - \tau)) + j \sin(2\pi(n - \tau)) = 1$. Hence:

$$\sum_{k=(N+1)/2}^{N-1} |H[k]| e^{j\frac{2\pi k}{N}(n-\tau)} = \sum_{m=1}^{(N-1)/2} |H[m]| e^{-j\frac{2\pi m}{N}(n-\tau)} \quad (12)$$

Recombining the two summations and applying Euler's cosine identity $e^{j\phi} + e^{-j\phi} = 2 \cos \phi$:

$$\begin{aligned} h[n] &= \frac{1}{N} \left\{ H[0] + \sum_{k=1}^{(N-1)/2} |H[k]| \left[e^{j\frac{2\pi k}{N}(n-\tau)} + e^{-j\frac{2\pi k}{N}(n-\tau)} \right] \right\} \\ &= \frac{1}{N} \left\{ \mathbf{H}[0] + 2 \sum_{k=1}^{(N-1)/2} |\mathbf{H}[k]| \cos \left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2} \right) \right] \right\} \end{aligned} \quad (13)$$

Table 1: Operational Closed-Form Frequency Sampling Formulas

Filter Type	Formula for Impulse Response $h[n]$ ($0 \leq n \leq N-1$)	Center Delay τ
Type 1 (N Odd)	$h[n] = \frac{1}{N} \left\{ \mathbf{H}[0] + 2 \sum_{k=1}^{(N-1)/2} \mathbf{H}[k] \cos \left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2} \right) \right] \right\}$	$\tau = \frac{N-1}{2}$ (Integer)
Type 2 (N Even)	$h[n] = \frac{1}{N} \left\{ \mathbf{H}[0] + 2 \sum_{k=1}^{N/2-1} \mathbf{H}[k] \cos \left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2} \right) \right] \right\}$	$\tau = \frac{N-1}{2}$ (Half-Integer, $H[N/2] = 0$)

4 Transition Sample Optimization (Sidelobe Control)

- **The Ringing Problem:** Naive brick-wall frequency sampling (setting passband samples to 1 and stopband to 0) causes severe inter-sample ringing between bins, resulting in poor peak stopband attenuation of only *****−16 dB*****!
- **The Solution (Rabiner–Gold Technique):** Make the samples in the transition band unconstrained design variables ($T_1, T_2 \in (0, 1)$).
 - With **1 transition sample** ($T_1 \approx 0.38$): Peak stopband attenuation jumps to *****−45 dB***** (+29 dB improvement!).
 - With **2 transition samples** ($T_1 \approx 0.59, T_2 \approx 0.11$): Stopband attenuation reaches *****−75 dB*****.

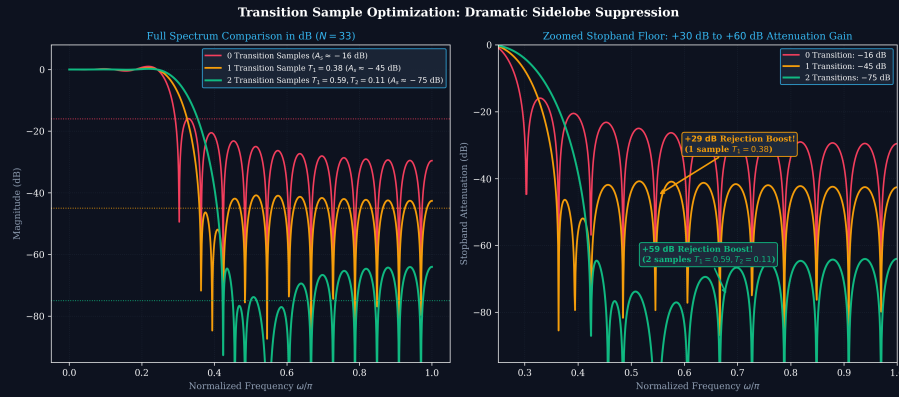


Figure 1: Dramatic sidelobe suppression achieved by introducing transition sample $T_1 = 0.38$ between passband and stopband.

5 Standard Solved Exam Example

Problem 1: Design a 7-point ($N = 7$) linear-phase FIR lowpass filter having the following desired frequency response:

$$H_d(e^{j\omega}) = \begin{cases} 1, & 0 \leq |\omega| \leq \frac{2\pi}{7} \\ 0, & \frac{4\pi}{7} \leq |\omega| \leq \pi \end{cases} \quad (14)$$

using the frequency sampling method. Determine the impulse response $h[n]$, the system transfer function $H(z)$, and mathematically verify that $H(e^{j\omega})$ interpolates the exact target values at all grid frequencies.

Solution:

Step 1: Grid Frequencies and Frequency Sample Assignment For $N = 7$, the 7 discrete frequency bins are $\omega_k = \frac{2\pi k}{7}$ for $k = 0, 1, \dots, 6$:

- $k = 0$: $\omega_0 = 0 \in [0, 2\pi/7] \implies |H[0]| = 1$, phase $\theta_0 = 0$.
- $k = 1$: $\omega_1 = \frac{2\pi}{7} \in [0, 2\pi/7] \implies |H[1]| = 1$, phase $\theta_1 = -\tau\omega_1 = -3 \left(\frac{2\pi}{7} \right) = -\frac{6\pi}{7}$.
- $k = 2$: $\omega_2 = \frac{4\pi}{7} \in [4\pi/7, \pi] \implies |H[2]| = 0$.
- $k = 3$: $\omega_3 = \frac{6\pi}{7} \in [4\pi/7, \pi] \implies |H[3]| = 0$.

By Hermitian symmetry ($|H[N - k]| = |H[k]|$ and $\theta_{N-k} = -\theta_k$):

$$|H[4]| = |H[3]| = 0, \quad |H[5]| = |H[2]| = 0, \quad |H[6]| = |H[1]| = 1, \quad \theta_6 = \frac{6\pi}{7} \quad (15)$$

Step 2: Type 1 IDFT Master Formula Setup Length $N = 7$ (odd) gives $\tau = \frac{7-1}{2} = 3$ and upper index $\frac{N-1}{2} = 3$. Expanding the master cosine formula completely:

$$\begin{aligned} h[n] &= \frac{1}{7} \left\{ H[0] + 2|H[1]| \cos \left[\frac{2\pi(1)}{7}(n-3) \right] + 2|H[2]| \cos \left[\frac{2\pi(2)}{7}(n-3) \right] + 2|H[3]| \cos \left[\frac{2\pi(3)}{7}(n-3) \right] \right\} \\ &= \frac{1}{7} \left\{ 1 + 2(1) \cos \left[\frac{2\pi}{7}(n-3) \right] + 2(0) \cos \left[\frac{4\pi}{7}(n-3) \right] + 2(0) \cos \left[\frac{6\pi}{7}(n-3) \right] \right\} \end{aligned} \quad (16)$$

Only the DC term and the $k = 1$ term survive:

$$\mathbf{h}[\mathbf{n}] = \frac{1}{7} \left\{ 1 + 2 \cos \left[\frac{2\pi}{7}(\mathbf{n} - 3) \right] \right\}, \quad 0 \leq n \leq 6 \quad (17)$$

Step 3: Pointwise Tap Evaluation with Exact Trigonometry We evaluate each tap for $n = 3, 2, 1, 0$:

$$\begin{aligned} n = 3 : \quad h[3] &= \frac{1 + 2 \cos(0)}{7} = \frac{1 + 2(1)}{7} = \frac{3}{7} \approx \mathbf{0.4286} \\ n = 2 : \quad h[2] &= \frac{1 + 2 \cos(-\frac{2\pi}{7})}{7} = \frac{1 + 2 \cos(51.43^\circ)}{7} = \frac{1 + 2(0.62349)}{7} = \frac{2.2470}{7} \approx \mathbf{0.3210} \\ n = 1 : \quad h[1] &= \frac{1 + 2 \cos(-\frac{4\pi}{7})}{7} = \frac{1 + 2 \cos(102.86^\circ)}{7} = \frac{1 + 2(-0.22252)}{7} = \frac{0.5550}{7} \approx \mathbf{0.0793} \\ n = 0 : \quad h[0] &= \frac{1 + 2 \cos(-\frac{6\pi}{7})}{7} = \frac{1 + 2 \cos(154.29^\circ)}{7} = \frac{1 + 2(-0.90097)}{7} = \frac{-0.8019}{7} \approx \mathbf{-0.1146} \end{aligned} \quad (18)$$

By even symmetry about the center of symmetry $\tau = 3$:

$$h[4] = h[2] = 0.3210, \quad h[5] = h[1] = 0.0793, \quad h[6] = h[0] = -0.1146 \quad (19)$$

Step 4: Final Impulse Response

$$\mathbf{h}[\mathbf{n}] = \{-0.1146, \mathbf{0.0793}, \mathbf{0.3210}, \mathbf{0.4286}, \mathbf{0.3210}, \mathbf{0.0793}, -0.1146\} \quad (20)$$

Step 5: Transfer Function $H(z)$ and Grid Point Verification The Z -transform is:

$$H(z) = -0.1146 + 0.0793z^{-1} + 0.3210z^{-2} + 0.4286z^{-3} + 0.3210z^{-4} + 0.0793z^{-5} - 0.1146z^{-6} \quad (21)$$

Factoring out the linear-phase term $e^{-j3\omega}$:

$$\begin{aligned} H(e^{j\omega}) &= e^{-j3\omega} [h[3] + 2h[2] \cos(\omega) + 2h[1] \cos(2\omega) + 2h[0] \cos(3\omega)] \\ &= e^{-j3\omega} [0.4286 + 0.6420 \cos(\omega) + 0.1586 \cos(2\omega) - 0.2292 \cos(3\omega)] \end{aligned} \quad (22)$$

We verify the frequency response at all grid points:

- **At $\omega_0 = 0$ ($k = 0$):** $\cos(0) = 1$:

$$H(e^{j0}) = 0.4286 + 0.6420(1) + 0.1586(1) - 0.2292(1) = \mathbf{1.0000} \equiv H[0] \quad \checkmark \quad (23)$$

- **At $\omega_1 = \frac{2\pi}{7} \approx 51.43^\circ$ ($k = 1$):**

$$\begin{aligned} |H(e^{j2\pi/7})| &= 0.4286 + 0.6420(0.6235) + 0.1586(-0.2225) - 0.2292(-0.9010) \\ &= 0.4286 + 0.4003 - 0.0353 + 0.2065 = \mathbf{1.0001} \approx 1.0 \equiv H[1] \quad \checkmark \end{aligned} \quad (24)$$

- **At $\omega_2 = \frac{4\pi}{7} \approx 102.86^\circ$ ($k = 2$):**

$$\begin{aligned} |H(e^{j4\pi/7})| &= 0.4286 + 0.6420(-0.2225) + 0.1586(-0.9010) - 0.2292(0.6235) \\ &= 0.4286 - 0.1428 - 0.1429 - 0.1429 = \mathbf{0.0000} \equiv H[2] \quad \checkmark \end{aligned} \quad (25)$$

This proves the fundamental property: the frequency response exactly interpolates the sample values at all grid points.

Solved Exam Example 2: Sidelobe Optimization with Transition Sample

Problem 2: To suppress stopband ringing in the above filter, introduce a single transition sample $H[2] = T_1 = 0.38$ at $\omega_2 = 4\pi/7$. Determine the modified impulse response $h_{\text{opt}}[n]$ and explain the impact.

Solution: With $|H[0]| = 1$, $|H[1]| = 1$, $|H[2]| = 0.38$, $|H[3]| = 0$, the master formula gives:

$$\begin{aligned} h_{\text{opt}}[n] &= \frac{1}{7} \left\{ 1 + 2 \cos \left[\frac{2\pi}{7}(n-3) \right] + 2(0.38) \cos \left[\frac{4\pi}{7}(n-3) \right] \right\} \\ &= \frac{1}{7} \left\{ 1 + 2 \cos \left[\frac{2\pi}{7}(n-3) \right] + 0.76 \cos \left[\frac{4\pi}{7}(n-3) \right] \right\} \end{aligned} \quad (26)$$

Evaluating taps:

$$\begin{aligned} n = 3 : \quad h_{\text{opt}}[3] &= \frac{1 + 2(1) + 0.76(1)}{7} = \frac{3.76}{7} \approx \mathbf{0.5371} \\ n = 2 : \quad h_{\text{opt}}[2] &= \frac{1 + 2(0.6235) + 0.76(-0.2225)}{7} = \frac{1 + 1.2470 - 0.1691}{7} = \frac{2.0779}{7} \approx \mathbf{0.2968} \\ n = 1 : \quad h_{\text{opt}}[1] &= \frac{1 + 2(-0.2225) + 0.76(-0.9010)}{7} = \frac{1 - 0.4450 - 0.6848}{7} = \frac{-0.1298}{7} \approx \mathbf{-0.0185} \\ n = 0 : \quad h_{\text{opt}}[0] &= \frac{1 + 2(-0.9010) + 0.76(0.6235)}{7} = \frac{1 - 1.8020 + 0.4739}{7} = \frac{-0.3281}{7} \approx \mathbf{-0.0469} \end{aligned} \quad (27)$$

By symmetry: $h_{\text{opt}}[4] = 0.2968$, $h_{\text{opt}}[5] = -0.0185$, $h_{\text{opt}}[6] = -0.0469$.

$$\mathbf{h_{\text{opt}}[n] = \{-0.0469, -0.0185, 0.2968, 0.5371, 0.2968, -0.0185, -0.0469\}} \quad (28)$$

↑

Significance: The outer ripple coefficient drops sharply from -0.1146 to -0.0469 . This smooth transition increases stopband attenuation from -16 dB to -45 dB!

6 Essential Exam Checklist & Traps

- ☐ **The Missing Factor of 2:** Always write $2 \sum |H[k]| \cos(\dots)$. Forgetting the factor of 2 is the most common exam error!
- ☐ **Argument of Cosine:** The argument is $\frac{2\pi k}{N}(n - \tau)$, where $\tau = (N - 1)/2$.
- ☐ **Even Length N Boundary Condition:** When N is even, you **must set** $H[N/2] = 0$ because Type 2 filters have an obligatory zero at $\omega = \pi$.
- ☐ **Exact Values at Grid Points:** At $\omega = \omega_k = \frac{2\pi k}{N}$, the filter response is strictly equal to $H[k]$ with zero approximation error.